



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY**

**Bachelor of Information Communication Technology
First Year - Semester II Examination - November 2023**

ICT 1202 – ELECTRONIC CIRCUITS

Time: Two (02) hours

- Answer **ALL** questions.
- There are **Four (04) essay questions** printed on **Three (03) pages**.

1)

i) Simplify the expression $Z = ABC + A\bar{B} \cdot (\bar{A} \bar{C})$

[4 Marks]

ii) A digital circuit takes four binary digits as an input and produces 1 as its output if the decimal value represented by the four binary digits is a **prime number** (number which can only be divided by itself and 1) and 0 otherwise. Assume that all four binary digits represent positive decimal values (No bit is allocated for the sign).

a) The following truth table is designed to describe the above circuit in which A, B, C and D represent the four binary inputs from the most significant bit to the least significant bit and F(A,B,C,D) as the output of the circuit. Copy the following truth table onto your answer sheet as it is and complete the output column.

[8 Marks]

A	B	C	D	F(A,B,C,D)
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

- b) Draw a Karnaugh-Map for the truth table in **part a** and get the simplified **Sum of Products** Boolean expression.

[8 Marks]

- c) Design the logic circuit for the expression you got in **part b**.

[5 Marks]

[Total 25 Marks]

2)

- i) List down combinational logic circuits related to data transmission. [4 Marks]
 ii) Using a truth table build the logic circuit for comparison of two 2-bit numbers P and Q. If require a HIGH output if two numbers are equal. [7 Marks]
 iii) Implement the following Boolean function with a multiplexer.
 $f(A,B,C,D) = \sum m(0,2,3,5,7,9,12,14)$ [10 Marks]
 iv) Describe two (02) applications of multiplexer. [4 Marks]

[Total 25 Marks]

3)

- i) Explain common base NPN configuration in a BJT Transistor with a suitable diagram. [5 Marks]
 ii) Describe the depletion region in a diode with an appropriate diagram. [5 Marks]
 iii) Find the current in each diode in the circuit shown below (Figure 3.1) using the constant voltage drop model with $V_{on} = 0.70\text{ V}$, $R_D = 0$. [7 Marks]

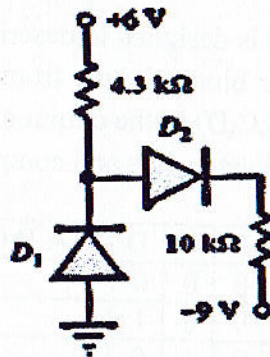


Figure 3.1

- iv) Find V_B , V_E and V_C when $R_B = 100\text{ k}\Omega$ and $\beta = 100$ in the transistor circuit given in the Figure 3.2. [8 Marks]

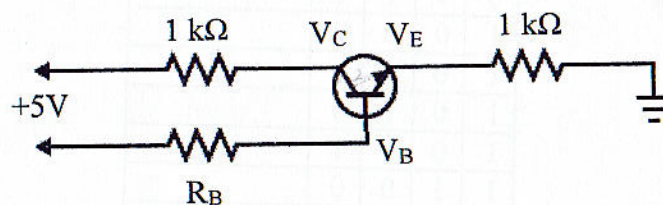


Figure 3.2

[Total 25 Marks]

4)

- i) What is P-type and N-type in semiconductors? [4 Marks]
 ii) Draw the diagram of Decimal to BCD Encoder build using OR gates. [7 Marks]
 iii) Find the equivalent resistance at the indicated terminal pair for each of the network shown in Figure 4.1. [5 Marks]

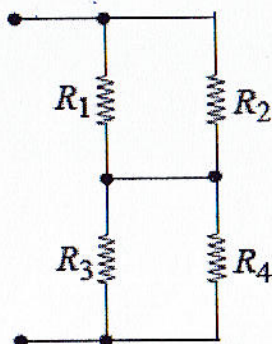


Figure 4.1

- iv) In the circuit shown below (Figure 4.2), the current through the $3\text{ k}\Omega$ resistor is measured to be 4 mA . Determine the source voltage and (V_s) and source current (I_s).

[9 Marks]

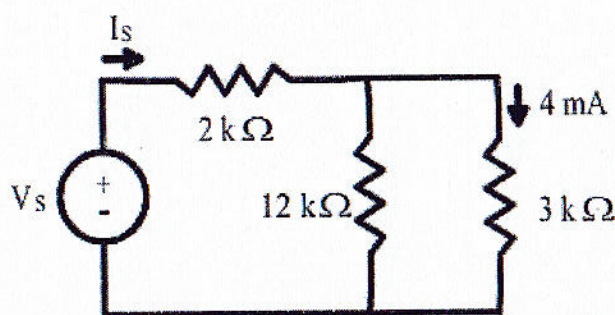


Figure 4.2

[Total 25 Marks]

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